Advanced Statistical Methods II
Lecture 19: Nonlinear Models and Least Squares

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Outline of Today's Lecture

The nonlinear regression model

The least-squares criterion

Exponential models

Compartmental models

Growth models

GLMs as nonlinear regression

The model and its assumptions

$$Y_i = f(\boldsymbol{x}_i, \boldsymbol{\theta}) + \varepsilon_i, \qquad i = 1, \ldots, n, \qquad \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p.$$

A1. **Correct mean:** $\mathbb{E}(Y_i \mid \boldsymbol{x}_i) = f(\boldsymbol{x}_i, \boldsymbol{\theta}_0)$ for some $\boldsymbol{\theta}_0 \in \Theta$, with f known.

A2. **Errors:** $\varepsilon_1, \ldots, \varepsilon_n$ independent, with mean 0 and common variance σ^2.

A3. **(Optional) Normality:** $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Needed only for exact likelihood statements.

Vector form: with $\boldsymbol{f}(\boldsymbol{\theta}) = \big(f(\boldsymbol{x}_1, \boldsymbol{\theta}), \ldots, f(\boldsymbol{x}_n, \boldsymbol{\theta})\big)^\top$,

$$\boldsymbol{y} = \boldsymbol{f}(\boldsymbol{\theta}) + \boldsymbol{\varepsilon}.$$

The nonlinear least squares estimator

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} S(\boldsymbol{\theta}), \qquad S(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})\}^2 = \|\boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})\|^2.$$

Differentiating, using $\partial \boldsymbol{f}/\partial \boldsymbol{\theta}^\top = \boldsymbol{J}$:

$$\nabla S(\boldsymbol{\theta}) = -2\, \boldsymbol{J}(\boldsymbol{\theta})^\top \{\boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})\},$$

so an interior minimizer satisfies the normal equations

$$\boldsymbol{J}(\widehat{\boldsymbol{\theta}})^\top \{\boldsymbol{y} - \boldsymbol{f}(\widehat{\boldsymbol{\theta}})\} = \boldsymbol{0}.$$

- Linear case: $\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \boldsymbol{0}$, which is linear in $\boldsymbol{\beta}$ and solved in closed form.
- Nonlinear case: p nonlinear equations in p unknowns. In general there is no closed form, and there may be several solutions.

NLS is maximum likelihood under normal errors

Under A1–A3, the log-likelihood is

$$\ell(\boldsymbol{\theta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{S(\boldsymbol{\theta})}{2\sigma^2}.$$

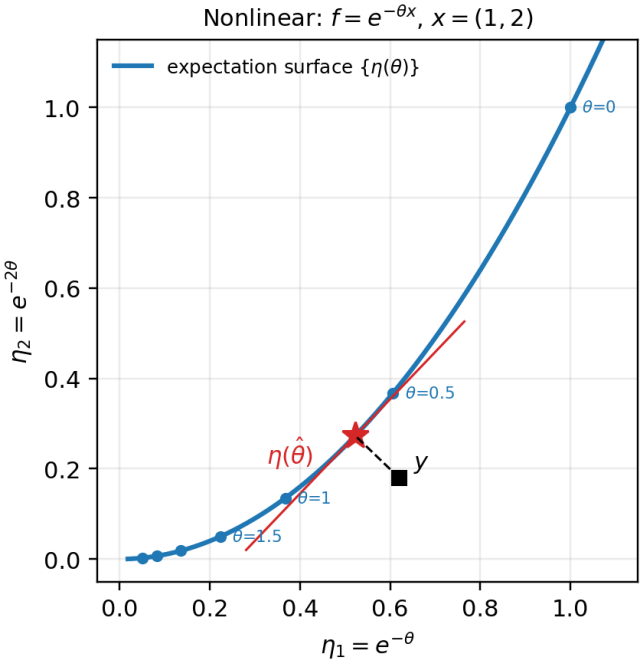
- For fixed $\boldsymbol{\theta}$, maximizing over σ^2 gives $\widehat{\sigma}^2(\boldsymbol{\theta}) = S(\boldsymbol{\theta})/n$.
- Substituting back gives the *profile* log-likelihood

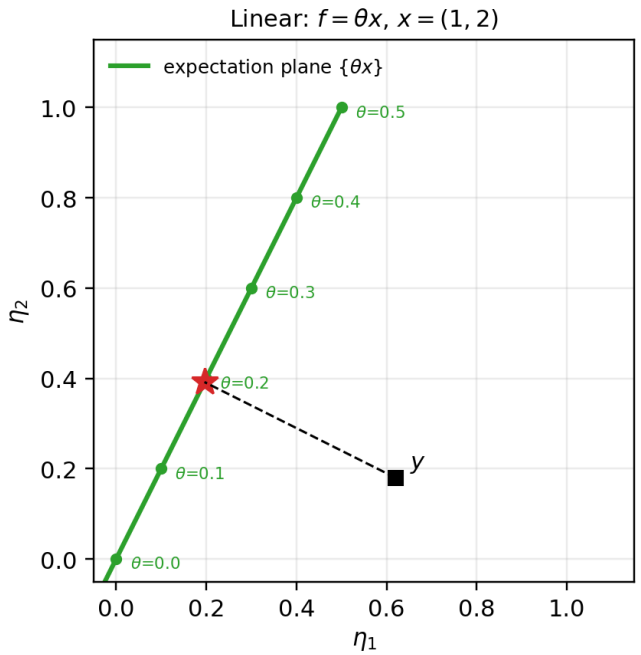
$$\ell_p(\boldsymbol{\theta}) = -\frac{n}{2}\log S(\boldsymbol{\theta}) + \text{const},$$

  which is a decreasing function of $S(\boldsymbol{\theta})$.
- Hence the MLE of $\boldsymbol{\theta}$ is the NLSE, and $\widehat{\sigma}^2_{\mathrm{ML}} = S(\widehat{\boldsymbol{\theta}})/n$.

In practice we use $s^2 = S(\widehat{\boldsymbol{\theta}})/(n-p)$, by analogy with linear regression.
It is no longer exactly unbiased, but it is the standard choice.

Geometry: the expectation surface





The set $\{\boldsymbol{f}(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\} \subset \mathbb{R}^n$ is the expectation surface (Bates and Watts, 1988). NLS finds the point on it closest to $\boldsymbol{y}$.

Reading the geometry

- **Normal equations = orthogonality.** $\boldsymbol{J}^\top(\boldsymbol{y} - \boldsymbol{f}(\widehat{\boldsymbol{\theta}})) = \boldsymbol{0}$ says the residual is perpendicular to the *tangent plane* at $\boldsymbol{f}(\widehat{\boldsymbol{\theta}})$. In the picture: $(\boldsymbol{y} - \widehat{\boldsymbol{f}})^\top \boldsymbol{t} = -6 \times 10^{-7}$, i.e. zero up to numerical tolerance.
- **Linear model:** the surface is a flat plane (the column space of $\boldsymbol{X}$), and equal steps in β give equal steps on the plane.
- **Nonlinear model, two departures:**
  1. the surface is *curved* (intrinsic nonlinearity);
  2. equal steps in θ give *unequal* steps along the surface (parameter-effects nonlinearity): see the tick marks $\theta = 0, 0.5, 1, \ldots$

Two routes to a nonlinear model

Mechanistic

Start from a law (mass balance, rate equation) and solve it.

- Parameters are physical: rates, capacities, volumes.
- Extrapolation is defensible *if* the law holds.
- Today: exponential, compartmental, growth.

Empirical

Choose a flexible curve with the right qualitative shape: bounded, monotone, S-shaped.

- Parameters describe features: asymptote, midpoint, slope.
- Interpolation is fine; extrapolation is risky.
- Example: Richards curves, the bounded model.

Many models start mechanistic and are then used empirically: the logistic curve began as a population law (Verhulst, 1838).

First-order kinetics

Law: the rate of change is proportional to the current amount,

$$\frac{dy}{dt} = -k\,y, \qquad y(0) = y_0.$$

Separating variables, $\int dy/y = -k \int dt$, gives

$$y(t) = y_0\, e^{-kt}.$$

- $k > 0$: decay (radioactivity, drug elimination). Half-life $t_{1/2} = \log 2/k$.
- $k < 0$: growth (compound interest, early epidemics). Doubling time $\log 2/|k|$.
- Regression model: $Y_i = \theta_1 e^{-\theta_2 t_i} + \varepsilon_i$, with $\boldsymbol{\theta} = (y_0, k)$.

$\log Y = \log \theta_1 - \theta_2 t$ is a linear model only if the error is *multiplicative*.

Non-linear models with an asymptote

Law: approach an equilibrium α at a rate proportional to the remaining distance,

$$\frac{dy}{dt} = k(\alpha - y), \qquad y(0) = y_0.$$

Put $u = \alpha - y$; then $u' = -ku$, so $u = (\alpha - y_0)e^{-kt}$ and

$$y(t) = \alpha - (\alpha - y_0)\, e^{-kt}.$$

- Newton's law of cooling; learning curves.

Compartmental models

Think of a system as a few well-mixed *boxes* (compartments), with material flowing between them.

- Inside a box, the material is evenly mixed: one number (the amount) describes the box.
- Material leaves a box at a rate proportional to how much is in it, like water draining from a tank: a fuller tank drains faster.
- Each arrow between boxes carries a rate constant: the fraction of the box's contents that moves per unit time.
- Bookkeeping for each box: *rate of change* = *inflow* − *outflow*. This gives one differential equation per box.

The same idea in many fields (Godfrey, 1983)

Pharmacology (gut → blood → elimination); epidemics (susceptible → infected → recovered, Kermack and McKendrick, 1927); radioactive decay chains.

An example

A patient swallows a tablet at time $t = 0$. What happens next?

Symbol	Meaning
D	the dose swallowed (here in mg per kg of body weight)
$A_g(t)$	amount of drug still in the **gut** (not yet absorbed) at time t; $A_g(0) = D$
k_a	**absorption** rate constant (per hour): drug moves from gut to blood at rate $k_a A_g(t)$
$A_b(t)$	amount of drug in the **blood** (the body) at time t; $A_b(0) = 0$
k_e	**elimination** rate constant (per hour): liver and kidneys remove drug at rate $k_e A_b(t)$
V	apparent volume of distribution (L/kg): links amount to concentration, $C(t) = A_b(t)/V$

- We never see the amounts. A blood sample gives the *concentration* $C(t)$.
- The curve tells a story: concentration rises while absorption beats elimination, peaks when they balance, then falls as elimination takes over.

The theophylline data

Theophylline is an asthma drug: too little does not work, too much is toxic, so dosing must be tailored. Data: a study by Dr. Robert Upton, reported by Boeckmann et al. (1994); R dataset `Theoph`. **12 subjects**, one oral dose each, serum concentration at **11 time points** over 25 hours (132 rows).

Variable	Meaning	Range
`Subject`	subject identifier	1–12
`Wt`	body weight (kg)	54.6–86.4
`Dose`	oral dose (mg/kg)	3.10–5.86
`Time`	hours since the dose	0–24.65
`conc`	concentration in the sample (mg/L)	0–11.4

- **Objects of interest:** each subject's k_a, k_e, $CL = k_e V$, hence $t_{1/2}$, $t_{\max}$, $C_{\max}$; and how they vary between subjects.

One compartment, oral dose: the equations



Mass balance (amounts, with dose D given at $t = 0$):

$$A_g' = -k_a A_g, \qquad A_b' = k_a A_g - k_e A_b, \qquad A_g(0) = D,\ A_b(0) = 0.$$

Step 1. The gut equation is first-order kinetics: $A_g(t) = De^{-k_a t}$.

Step 2. The blood equation becomes $A_b' + k_e A_b = k_a D e^{-k_a t}$. Multiply by the integrating factor $e^{k_e t}$:

$$\frac{d}{dt}\left(e^{k_e t} A_b\right) = k_a D\, e^{(k_e - k_a)t}.$$

One compartment, oral dose: the solution

Step 3. Integrate from 0 to t (with $k_a \neq k_e$):

$$A_b(t) = \frac{k_a D}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right).$$

Step 4. Concentration = amount / volume of distribution V:

$$\boxed{C(t) = \frac{D\,k_a}{V(k_a - k_e)}\left(e^{-k_e t} - e^{-k_a t}\right)} \qquad \text{(Bateman function)}$$

Related models:

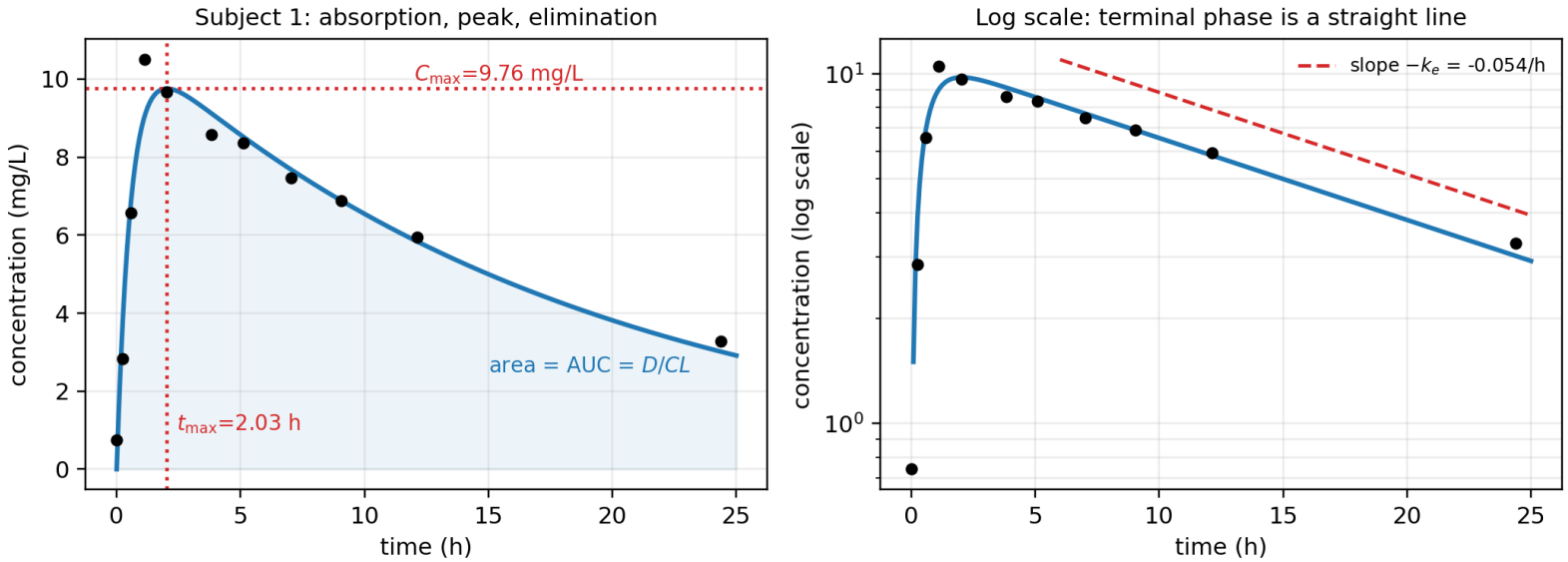
- IV bolus (drug is injected directly into a vein, so it enters the bloodstream immediately, with no gut to pass through. No absorption phase): $C(t) = (D/V)e^{-k_e t}$, simply exponential decay.
- Two compartments (blood and tissue): $C(t) = Ae^{-\alpha t} + Be^{-\beta t}$, a sum of exponentials.

Quantities clinicians use

All of these are functions of (k_a, k_e, V):

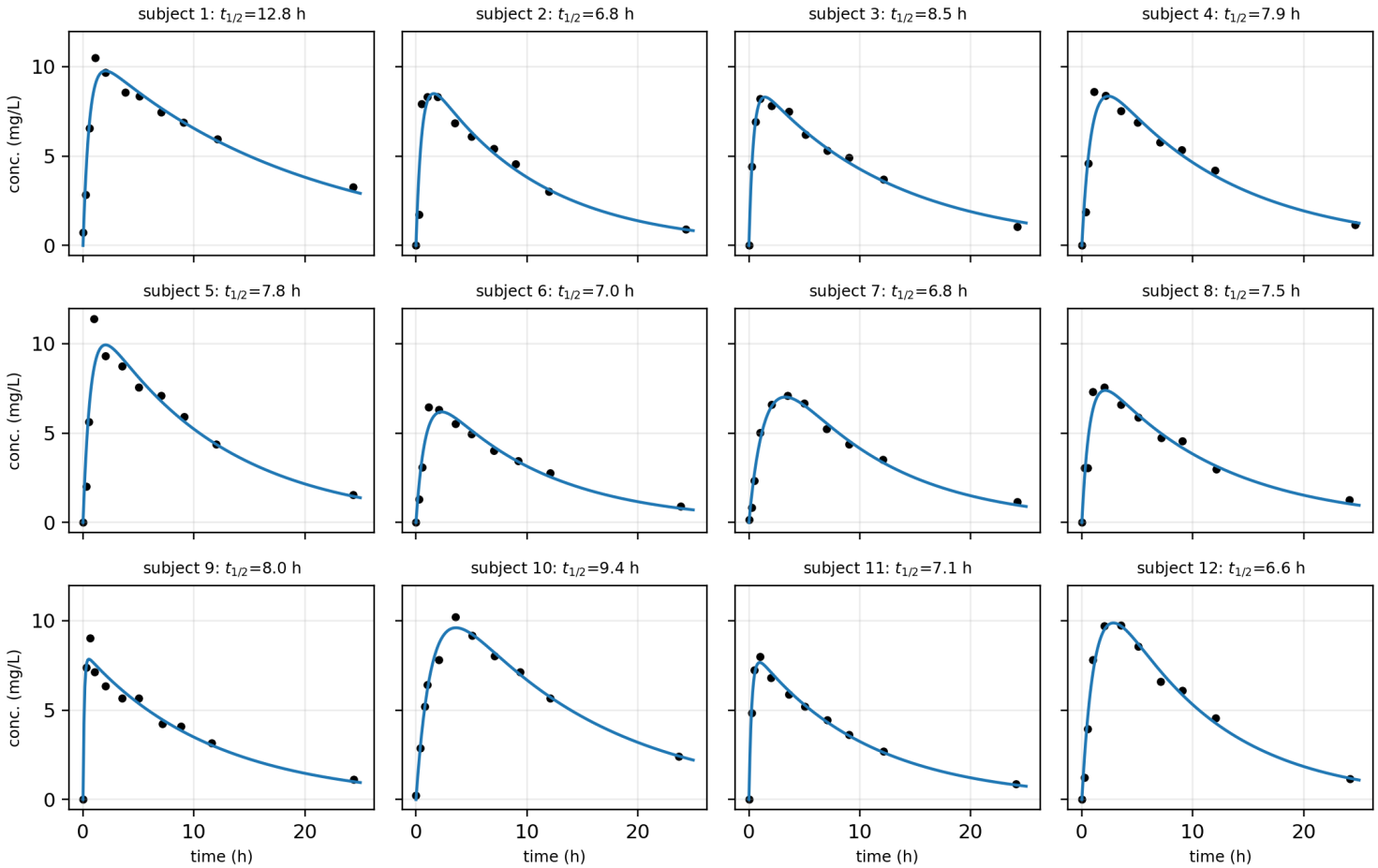
- **Elimination half-life:** $t_{1/2} = \log 2 / k_e$.
- **Clearance:** $CL = k_e V$ (volume of plasma cleared per unit time).
- **Exposure:**
$$\mathrm{AUC} = \int_0^\infty C(t)\,dt = \frac{Dk_a}{V(k_a - k_e)}\left(\frac{1}{k_e} - \frac{1}{k_a}\right) = \frac{D}{Vk_e} = \frac{D}{CL}.$$
- **Time of peak:**
$$C'(t) = 0 \iff k_e e^{-k_e t} = k_a e^{-k_a t} \iff t_{\max} = \frac{\log(k_a/k_e)}{k_a - k_e}.$$
- **Peak concentration:** $C_{\max} = C(t_{\max})$.

Theophylline, subject 1: anatomy of the fit



- Fit: $\widehat{k_a} = 1.78/\text{h}$, $\widehat{k_e} = 0.054/\text{h}$, $\widehat{CL} = 0.020$ L/(h·kg),
- On the log scale, the terminal phase is a straight line with slope $-k_e$.

All twelve subjects



One model and twelve sets of parameters. Each fit uses only 11 points to estimate 3 parameters.

Between-subject variability

Subj.	$\widehat{k_a}$ (1/h)	$\widehat{k_e}$ (1/h)	$\widehat{CL}$ (L/h/kg)	$\widehat{V}$ (L/kg)	$t_{1/2}$ (h)	$t_{\max}$ (h)
1	1.78	0.054	0.020	0.37	12.8	2.03
2	1.94	0.102	0.045	0.44	6.8	1.60
5	1.47	0.088	0.044	0.49	7.8	2.03
7	0.68	0.102	0.052	0.51	6.8	3.28
9	8.87	0.087	0.033	0.38	8.0	0.53
10	0.70	0.074	0.032	0.44	9.4	3.61

- Absorption varies over a factor of 13 (0.68 to 8.87/h); half-life varies from 6.6 to 12.8 h across all 12 subjects.
- Subject 9 absorbs so fast that only one or two points lie on the rising limb, so $\widehat{k_a}$ is poorly determined.
- Modelling all subjects jointly, with random effects, is the subject of nonlinear mixed models (Pinheiro and Bates, 2000). That is beyond this course.

A built-in identifiability problem: flip-flop

Swap the roles of k_a and k_e:

$$\frac{Dk_e}{V'(k_e - k_a)}\left(e^{-k_a t} - e^{-k_e t}\right) = \frac{Dk_e}{V'(k_a - k_e)}\left(e^{-k_e t} - e^{-k_a t}\right).$$

This equals the original $C(t)$ when $V' = Vk_e/k_a$.

- So (k_a, k_e, V) and $(k_e, k_a, Vk_e/k_a)$ give **identical** curves.
- Concentration data alone cannot tell absorption from elimination. The convention $k_a > k_e$ resolves it.
- An algorithm may converge to either solution, depending on where it starts.

This is *structural* non-identifiability (Lecture 17): no amount of data of this type removes it. We restrict Θ to break the symmetry.

The logistic model

Law (Verhulst, 1838): growth proportional to size and to the remaining room,

$$\frac{dy}{dt} = r\,y\left(1 - \frac{y}{K}\right).$$

Partial fractions, $\dfrac{K}{y(K-y)} = \dfrac{1}{y} + \dfrac{1}{K-y}$, give $\log \dfrac{y}{K-y} = r(t-m)$, so

$$y(t) = \frac{K}{1 + e^{-(t-m)/s}}, \qquad s = 1/r.$$

- K: upper asymptote (carrying capacity, adult size).
- m: time at which $y = K/2$. This is the inflection point, where growth is fastest.

Gompertz, Richards, von Bertalanffy

Model	Rate law	Solution
Gompertz	$y' = ry\log(K/y)$	$K\exp\{-be^{-rt}\}$
Richards	$y' = \frac{r}{\nu}y\{1-(y/K)^{\nu}\}$	$K\{1+be^{-rt}\}^{-1/\nu}$
von Bertalanffy	$W' = aW^{2/3} - cW$	$W_\infty(1-be^{-kt})^3$

Sources: Gompertz (1825), Richards (1959), von Bertalanffy (1938).

- The Gompertz inflection is at $y = K/e \approx 0.37K$: early, with a long slow approach to K. It is used for tumours and mortality.
- Richards nests both: $\nu = 1$ is logistic, and $\nu \to 0$ gives Gompertz. The inflection is at $K(1+\nu)^{-1/\nu}$.
- von Bertalanffy: anabolism scales like surface area ($W^{2/3}$), catabolism like mass. Length $L \propto W^{1/3}$ then follows asymptotic regression, $L_\infty(1-e^{-k(t-t_0)})$.

How are these growth models related?

Gompertz and von Bertalanffy are not special cases of the logistic. They are different models. But all of them are members of one family, the Richards curve (Richards, 1959):

$$\frac{dy}{dt} = \frac{r}{\nu} y\left\{1 - \left(\frac{y}{K}\right)^{\nu}\right\}, \qquad y(t) = K\{1 + be^{-rt}\}^{-1/\nu}.$$

Shape ν	Model	Fastest growth at size
$\nu = 1$	logistic	$K/2 = 0.50K$
$\nu \to 0$ (limit)	Gompertz	$K/e \approx 0.37K$
$\nu = -1/3$ (with $b < 0$)	von Bertalanffy (weight)	$(2/3)^3 K \approx 0.30K$

- In general, the inflection is at $y = K(1+\nu)^{-1/\nu}$ (for $\nu > -1$). The shape parameter ν decides *where* growth is fastest.
- Check: with $\nu = -1$ the rate law becomes $y' = r(K - y)$, the asymptotic regression law from earlier.

The orange tree data

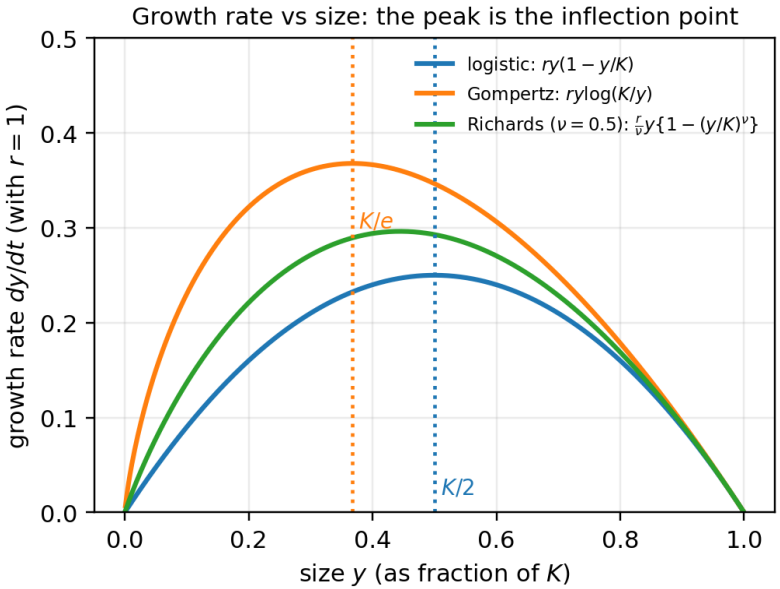
- From Draper and Smith (1998); available in R as `Orange`.
- **5 trees**, each measured on the same **7 occasions**: 35 rows.

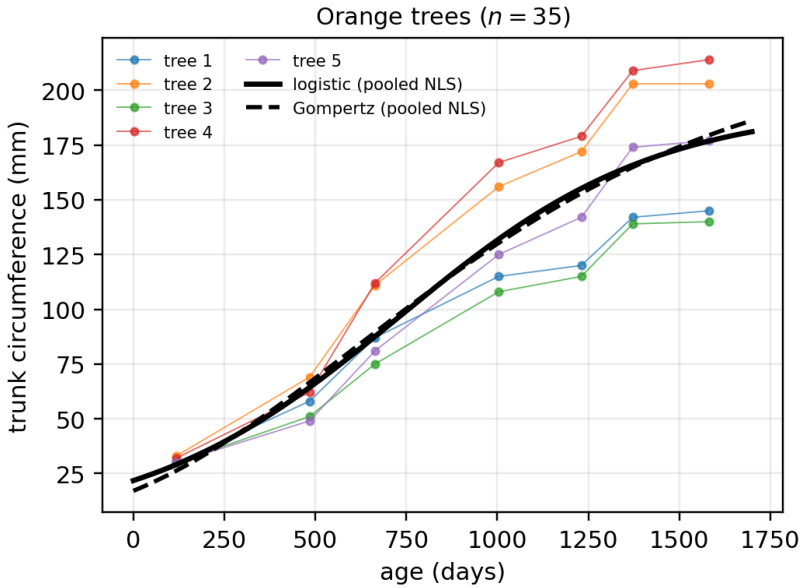
Variable	Meaning	Values
`Tree`	which tree	1–5
age	age in days (from 31 Dec 1968)	118, 484, 664, 1004, 1231, 1372, 1582
`circumference`	trunk circumference (mm)	30–214

- Circumference is probably taken at breast height (standard in forestry). Largest values by tree: $145, 203, 140, 214, 177$ mm: the trees clearly differ.
- **Objects of interest:** the final size (asymptote), the age of fastest growth (inflection), and the speed of growth.

Repeated measurements on the same tree are not independent (assumption A2). Today we pool the trees for simplicity; tree-specific (mixed) models are beyond this course.

Growth models: picture and data





- Left: the rate law decides *where* growth is fastest: $K/2$ (logistic), K/e (Gompertz), $0.44K$ (Richards, $\nu = 0.5$).
- Right: trunk circumference of five orange trees (Draper and Smith, 1998; Pinheiro and Bates, 2000).

Where ordinary regression struggles

Three very common kinds of response:

- **Yes/no** (binary): did the O-ring show distress? Did the patient recover?
- **Proportions**: k successes out of m trials, e.g. 2 of 6 O-rings.
- **Counts**: $0, 1, 2, \ldots$ accidents, insurance claims, disease cases.

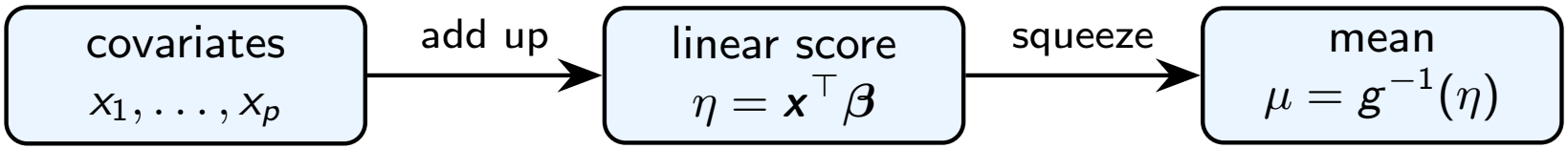
A straight line through the Challenger proportions

Fitting (proportion distressed) $= a + b \times$ temperature by OLS gives a line that reaches 0 at 77.8°F and predicts -0.025 at 81°F. That is a *negative probability*, at a temperature where a launch actually took place.

Two problems, not one:

1. **Range:** probabilities live in $[0, 1]$, counts in $[0, \infty)$; a straight line ignores this.
2. **Spread:** the variance changes with the mean. A proportion near 0 or 1 varies less than one near $1/2$; large counts vary more than small counts.

The GLM idea in plain words



1. **Choose a distribution** that matches the type of response: binomial for "k out of m", Poisson for counts, normal for measurements.
2. **Build a score** from the covariates, exactly as in linear regression: $\eta = \beta_0 + \beta_1 x_1 + \cdots$. It can be any real number.
3. **Squeeze the score into the allowed range** with an inverse link g^{-1}: an S-curve for probabilities, the exponential for counts.

Linear regression is the special case: normal distribution, no squeezing ($\mu = \eta$). Everything else is the same recipe with a different distribution and a different squeeze.

Logistic regression: reading the coefficients

For a probability π, model the log-odds linearly:

$$\log \frac{\pi}{1-\pi} = \beta_0 + \beta_1 x \quad \Longleftrightarrow \quad \pi = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}.$$

- **Odds** $= \pi/(1-\pi)$: e.g. $\pi = 0.2$ means odds of $1:4$.
- Increasing x by one unit *multiplies the odds* by e^{β_1}, the odds ratio.
- On the probability scale the effect is not constant: $d\pi/dx = \beta_1 \pi(1-\pi)$, largest at $\pi = 1/2$ and tiny near 0 or 1.

Challenger: $\widehat{\beta}_1 = -0.1156$ per °F

$e^{-0.1156} = 0.89$: each degree warmer multiplies the odds of distress by 0.89 (about 11% lower). Ten degrees warmer: odds $\times 0.31$. Ten degrees *colder*: odds $\times 3.2$.

Poisson regression: counts

For a count Y with mean μ, model the log of the mean linearly:

$$\log \mu = \beta_0 + \beta_1 x \quad \Longleftrightarrow \quad \mu = e^{\beta_0} \left(e^{\beta_1}\right)^x.$$

- The mean is always positive, whatever the covariates.
- Effects are **multiplicative**: one unit more of x multiplies the expected count by e^{β_1} (a rate ratio).
- The Poisson variance equals the mean, $\mathrm{Var}(Y) = \mu$: bigger counts are noisier.
- Different observation periods or population sizes t_i enter as $\log \mu_i = \log t_i + \boldsymbol{x}_i^\top \boldsymbol{\beta}$ (an *offset*).

The mean $e^{\beta_0 + \beta_1 x}$ is exactly the exponential growth model from earlier in this lecture. What differs is the assumed *variance*, and therefore how the parameters should be estimated.

Fitting a GLM: trust precise data more

- GLMs are fitted by **maximum likelihood**, not by ordinary least squares.
- Intuition: an observation with small variance carries more information, so it should count more.
- For a proportion out of m trials, $\mathrm{Var} = \pi(1-\pi)/m$:

π	0.05	0.2	0.5
SD of a proportion out of $m = 6$	0.089	0.163	0.204

- Maximum likelihood turns out to be a *weighted* least-squares problem, with weights that depend on the unknown $\boldsymbol{\beta}$ (through π or μ).
- So it is solved by repeatedly re-weighting and re-fitting: iteratively reweighted least squares (Lecture 19).

GLM = a nonlinear mean + a variance that depends on the mean + a likelihood that accounts for both. The next two slides make this precise.

Exponential Family

An exponential-family density can be written

$$f(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{\phi} + c(y,\phi)\right\},$$

and a standard calculation gives $\mathbb{E}\, Y = b'(\theta) = \mu$ and Var $Y = \phi\, b''(\theta)$. Since θ is determined by μ, the second derivative $b''(\theta)$ can be written as a function of μ; that function is $V(\mu)$

Generalized linear models: the structure

A GLM (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989) has three parts:

1. **Random part:** Y_i from an exponential family, with $\mathbb{E}\, Y_i = \mu_i$ and $\operatorname{Var} Y_i = \phi V(\mu_i)$.
2. **Linear predictor:** $\eta_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}$.
3. **Link:** $g(\mu_i) = \eta_i$, so the mean function is $\mu_i = g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta})$.

Family	canonical link $g(\mu)$	mean function	$V(\mu)$
Normal	μ	$\boldsymbol{x}^\top \boldsymbol{\beta}$	1
Poisson	$\log \mu$	$e^{\boldsymbol{x}^\top \boldsymbol{\beta}}$	μ
Binomial (m trials)	$\log \frac{\pi}{1-\pi}$	$m/(1 + e^{-\boldsymbol{x}^\top \boldsymbol{\beta}})$	$m\pi(1-\pi)$

The mean is *nonlinear in* $\boldsymbol{\beta}$, so a GLM is a nonlinear regression model. But the variance changes with the mean, so unweighted least squares is not the right criterion.

Score equations: GLM versus NLS

NLS normal equations (constant variance):

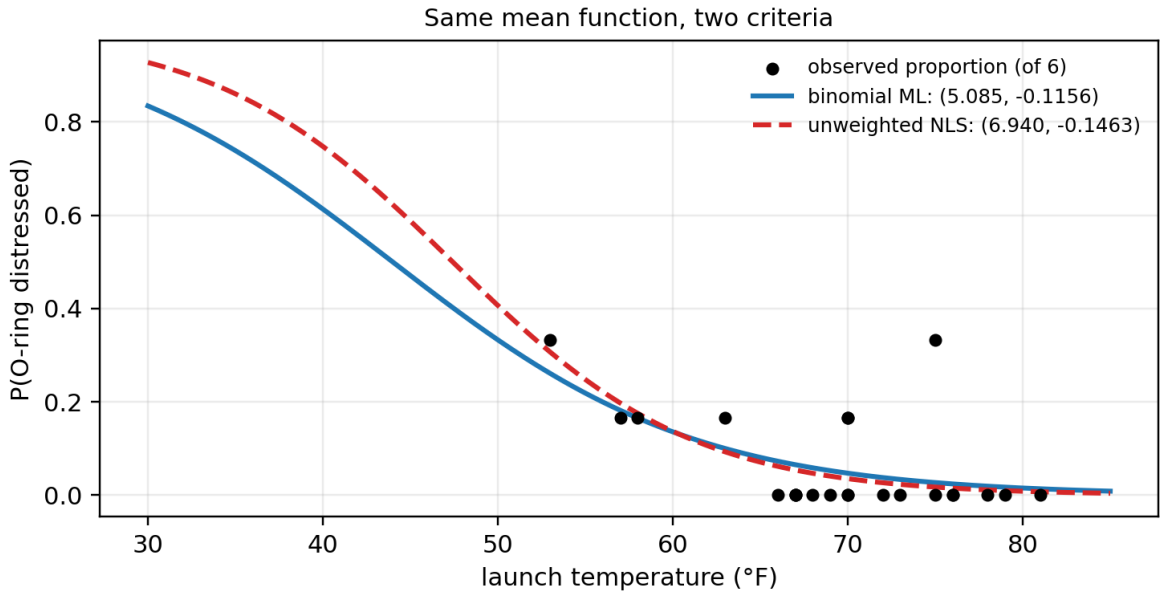
$$\sum_{i=1}^{n} \{y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})\} \frac{\partial f(\boldsymbol{x}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{0}.$$

GLM likelihood equations:

$$\sum_{i=1}^{n} \frac{y_i - \mu_i(\boldsymbol{\beta})}{\phi\, V(\mu_i)} \frac{\partial \mu_i(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \mathbf{0}.$$

- The same structure: residual $\times$ sensitivity, with an extra weight $1/V(\mu_i)$.
- The weights depend on $\boldsymbol{\beta}$, so the GLM fit is an *iteratively reweighted* NLS. This is Fisher scoring, in Lecture 19.
- Binomial with logit link: $\partial \mu_i / \partial \boldsymbol{\beta} = m\pi_i(1-\pi_i)\boldsymbol{x}_i = V(\mu_i)\boldsymbol{x}_i$, so the equations reduce to $\sum_i (k_i - m\pi_i)\boldsymbol{x}_i = \mathbf{0}$.

Challenger again: the criterion matters



- Same logistic mean function, two criteria. The binomial ML estimate is $(5.085, -0.1156)$ (Dalal et al., 1989); unweighted NLS on the proportions gives $(6.940, -0.1463)$.
- At 31°F: $\hat{\pi} = 0.82$ (ML) versus 0.92 (NLS). Unweighted NLS treats a proportion near 0 as just as noisy as one near $1/2$, which it is not.

Challenger: what should we conclude?

1. Same data, same S-shaped curve, two different answers. The slope is -0.146 (NLS) versus -0.116 (ML). How you measure "fit" is part of the method.

2. Why they differ: who gets a say. The two sets of estimating equations differ by one factor:

$$\text{ML:} \quad \sum_i (p_i - \pi_i)\, \boldsymbol{x}_i = \mathbf{0}, \qquad \text{unweighted NLS:} \quad \sum_i (p_i - \pi_i)\, \underbrace{\pi_i(1-\pi_i)}_{\text{extra factor}}\, \boldsymbol{x}_i = \mathbf{0},$$

where $p_i = k_i/6$. Most flights were warm, with $\pi_i \approx 0$; NLS multiplies their residuals by the small factor $\pi_i(1-\pi_i)$, shrinking their say and letting the few cold, damaged flights steer the curve: hence a steeper curve.

3. Which is right? ML, because it uses the binomial variance: a proportion near 0 really is more precise (SD 0.089 at $\pi = 0.05$, versus 0.204 at $\pi = 0.5$).

4. What survives either way: both say the risk per O-ring at 31°F is very high (0.82 or 0.92). But both are *extrapolations*: the coldest previous launch was 53°F.

When the variance changes with the mean, use the likelihood (a GLM) or appropriate weights, not unweighted least squares.

Key takeaway messages

1. The model is $Y = f(\boldsymbol{x}, \boldsymbol{\theta}) + \varepsilon$; the Jacobian $\boldsymbol{J}(\boldsymbol{\theta})$ plays the role of $\boldsymbol{X}$.
2. Mechanisms give models: first-order kinetics give exponentials; mass balance gives the Bateman function; rate laws give logistic, Gompertz, and Richards curves.
3. Parameters carry meaning, and derived quantities ($t_{1/2}$, AUC, $t_{\max}$) are functions of them.
4. GLMs are nonlinear regressions whose natural criterion is a reweighted one.
5. NLSE = MLE under normal errors. Geometrically, it is the projection of $\boldsymbol{y}$ onto a curved surface.
6. Existence and uniqueness are not automatic: watch for flip-flop, multiple minima, and flat valleys.

Exercises

1. Solve $y' = k(\alpha - y)$ with $y(0) = y_0$, and show that α, y_0, and $\log k$ are the parameters of `SSasymp`.
2. For the Bateman function, verify $\mathrm{AUC} = D/CL$ and $t_{\max} = \log(k_a/k_e)/(k_a - k_e)$. What happens to $t_{\max}$ as $k_a \to \infty$?
3. Fit the three-parameter `SSasymp` model to the BOD data, and compare it with the two-parameter $\theta_1(1 - e^{-\theta_2 x})$. Look at the estimate and standard error of `R0`. How many parameters can 6 observations support?
4. Show that the logistic curve's inflection point is at $y = K/2$ and the Gompertz curve's at $y = K/e$.
5. Derive the binomial-logit likelihood equations $\sum_i (k_i - m\pi_i)\boldsymbol{x}_i = \mathbf{0}$ from the general GLM score equations.
6. Give your own example, from your field, of a model with a structural identifiability problem, and suggest a constraint that removes it.

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